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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CSE) (CBCS) Degree Examinations
May – June 2025

23CSE233 – SOFTWARE ENGINEERING AND MODELLING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each**
and **One full** question from **Unit – III.**

Unit – I		Marks	BT*	CO*	PO*
1.	a) Construct V model for software development and explain.	10	L*3	1	2
	b) Justify the need for prototyping ? When prototyping can best adopted.	10	L3	1	2
2.	a) Summarize the agile methodology principles.	10	L2	1	1
	b) Show different architecture styles with diagrams.	10	L2	2	1
3.	a) Show the all types non functional software requirements and explain.	10	L2	2	1
	b) Illustrate pipes and filter architecture diagram.	10	L2	2	1
Unit – II					
4.	a) Outline Object-Oriented Modeling and Design (OOMD) principles with examples.	10	L2	3	1
	b) What is a design pattern? Why is it required? Explain different types of design patterns.	10	L2	3	1
5.	a) Summarize the various levels of software testing.	10	L2	3	1
	b) Illustrate Black box testing design techniques.	10	L2	3	1
6.	a) What is the Static Review process? Discuss Tools used for reviewing.	10	L2	4	1
	b) Explain the importance, key objectives and types of Software Maintenance .	10	L2	4	1
Unit – III					
7.	a) Interpret Reuse based Software Engineering Approaches.	10	L2	5	1
	b) Discuss the Key Features, challenges and benefits of COTS Systems.	10	L2	5	1
8.	a) Explain the types, characteristics life cycle of Component-Based Software Engineering (CBSE) Components.	10	L2	5	1
	b) Illustrate Key Features, Characteristics and Advantages of Distributed Software Engineering.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
